

## 04. NOTIONS OF EPIGENETICS, TENSEGRITY, ELECTROMAGNETISM

DURATION : 06:25

STUDY SHEET

### COURSE SUMMARY

This video explores the concepts of epigenetics, tensegrity, and electromagnetism, revealing how the environment modulates our genetic code. As osteopaths, it is crucial to understand that our actions and interactions can alter cellular behaviour and transmit transgenerational information. The concept of tensegrity reminds us that every touch is an integral vibration, influenced by various fields and stimuli. Together, these ideas provide an essential framework for a more conscious and holistic therapeutic approach.

## NOTIONS OF EPIGENETICS, TENSEGRITY AND ELECTROMAGNETISM

### THE IMPACT OF ENVIRONMENT ON GENETICS

Initially, our understanding of **genetics** led us to believe that everything was predetermined. It was thought that **DNA** contained all the necessary information to generate an organism. However, research has revealed a surprising truth: only **5 to 10%** of our genes are actually used, the rest having long been considered "useless".

### EPIGENETICS: WHEN ENVIRONMENT REDEFINES THE CODE

With advancing knowledge, we have discovered the colossal impact of the **environment** on genetic behaviour. All these environmental systems can modify this behaviour at an ultra-rapid pace.

Imagine a DNA strand, composed of molecules and amino acids forming **genetic codons** that enable protein production. This system is not static; it continuously modifies and adapts. A change in environment, an emotional shock, or a powerful event can alter this structure.

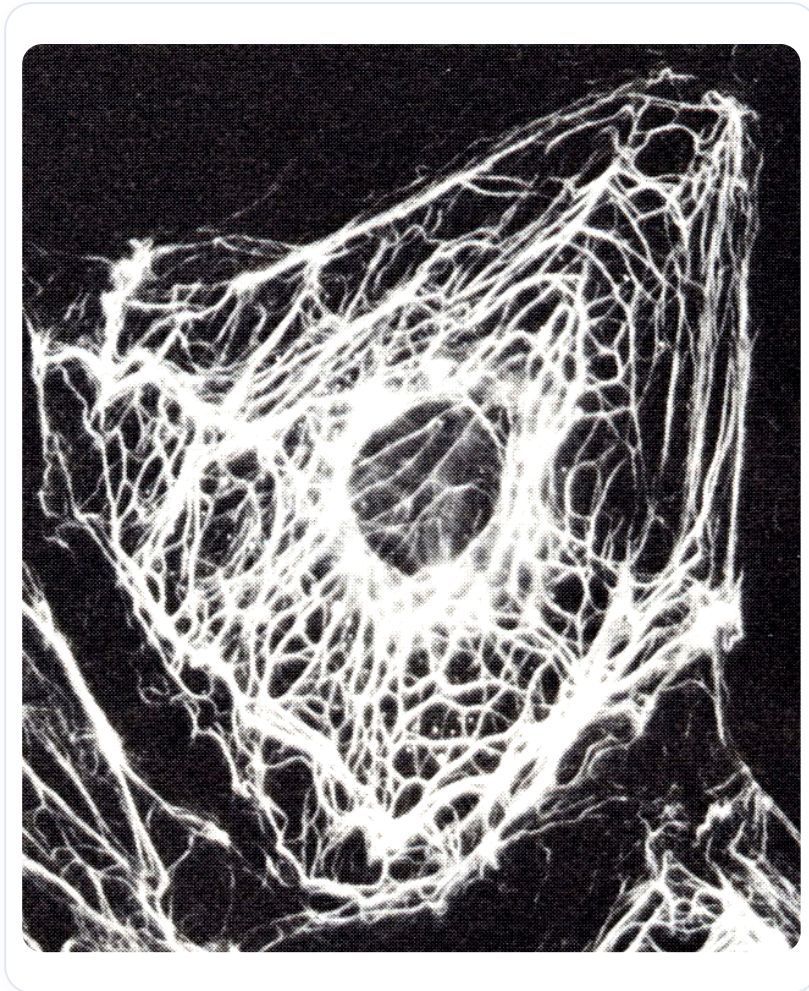
### OUR THERAPEUTIC RESPONSIBILITY

As **osteopaths**, our knowledge of anatomy and its function allows us to realise the profound impact of embryology. As Anthony Schiller states, this impact extends to the cellular level and even beyond. We have a responsibility: to be mindful of our actions and how we touch. Every

interaction has profound repercussions.

## THE CELL: A UNIVERSE OF TENSEGRITY

Let's take the example of a **cell**. It is connected to other cells by **intercellular adhesion molecules**, which maintain the structure. What we have discovered is that the cell possesses a **cytoskeleton**. This internal skeleton is connected to the membrane and extends to the nucleus, forming lines of force and precise structures within the cell.



*FIG. — Cell cytoskeleton*

## THE CONCEPT OF TENSEGRITY

This is where the fundamental concept of **tensegrity** in osteopathy comes in. If I touch someone, I touch the entirety of their body. It is a frequency, a vibration that propagates throughout the entire organism.

Let's observe a cell with its **extracellular matrix** and the junction between the cell and its external environment. What happens at this junction contains profound information.

## ENVIRONMENTAL INFLUENCES ON THE CELL

An **electric field**, an **electromagnetic field**, light, temperature, pH, mechanical stress, shocks, deformations, and hydrostatic pressures – all of these influence cellular behaviour. This is not a small impact, but a global environment that acts on the system.

## TRANSGENERATIONAL INHERITANCE AND FUNDAMENTAL FORCES

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We have also discovered that our genetic programme carries **transgenerational** impact in our behaviours.

### THE IMPRINT OF PAST GENERATIONS

Traumas experienced by our ancestors can be carried and manifest. It has been observed that we can carry information in our genetic code for up to **three generations**, sometimes even more. We are sometimes here to release this information.

Genetic studies have been able to trace back up to three generations to identify behaviours or appearances that resurface. This suggests that our technique should not be a simple manipulation, but a frequency, a vibration that interacts with this profound information.

### THE FOUR FUNDAMENTAL FORCES OF LIFE

The presence of elementary particles implies forces in life. We currently recognise four major forces:

- **The weak force**
- **The strong force**
- **The electromagnetic force**
- **The force of gravity**

The weak and strong forces are nuclear forces that are beyond us. To give you an image, if one of these forces were removed from the universe, everything would collapse like a house of cards.

### THE IMPORTANCE OF THE ELECTROMAGNETIC FORCE

The **electromagnetic force** is particularly interesting because it allows for the interactions of systems. It is responsible for bonding and allows matter to exist in its form. The organism must therefore involve and possess systems to call upon these forces. Where does the electromagnetic force come from? It is there before us. When matter appears, it must have this capacity.

## THE POLARISED HUMAN AND MAXWELL'S LAW

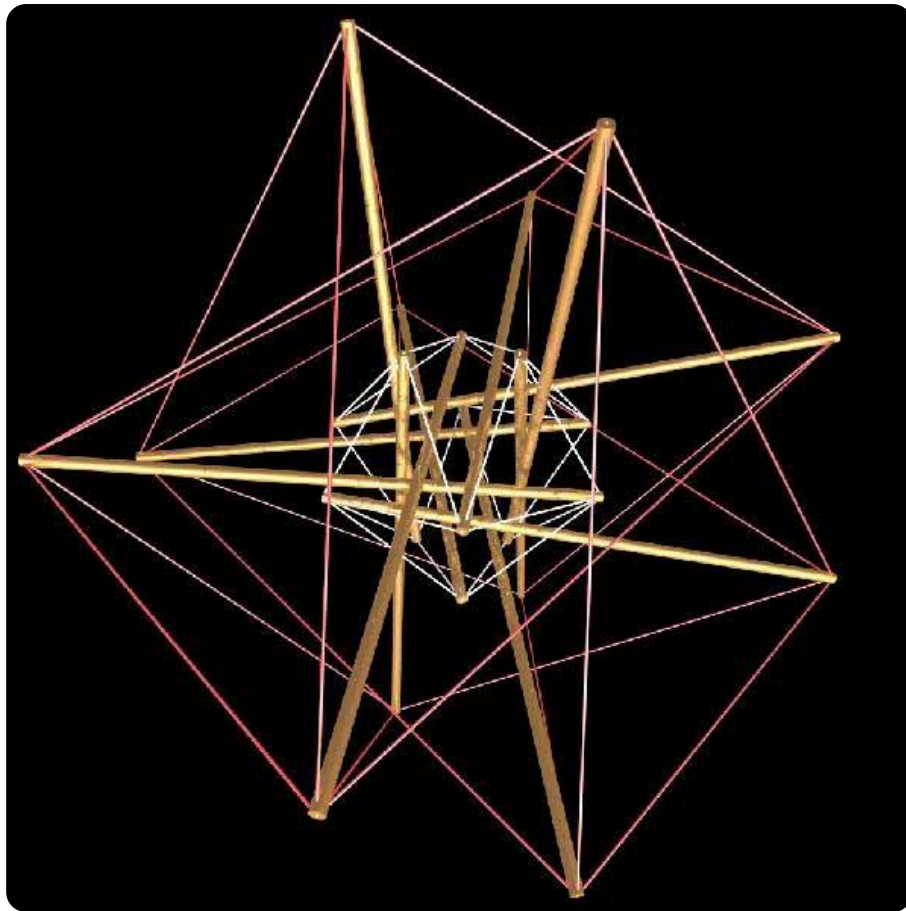
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This brings us to the concept of the **polarised human**. **Maxwell's Law** is very simple: if you have an electric field, you always have an electromagnetic field around it.

In other words, an electric field inevitably generates a surrounding electromagnetic field. This interaction is fundamental to understanding how our body functions and interacts with its environment.

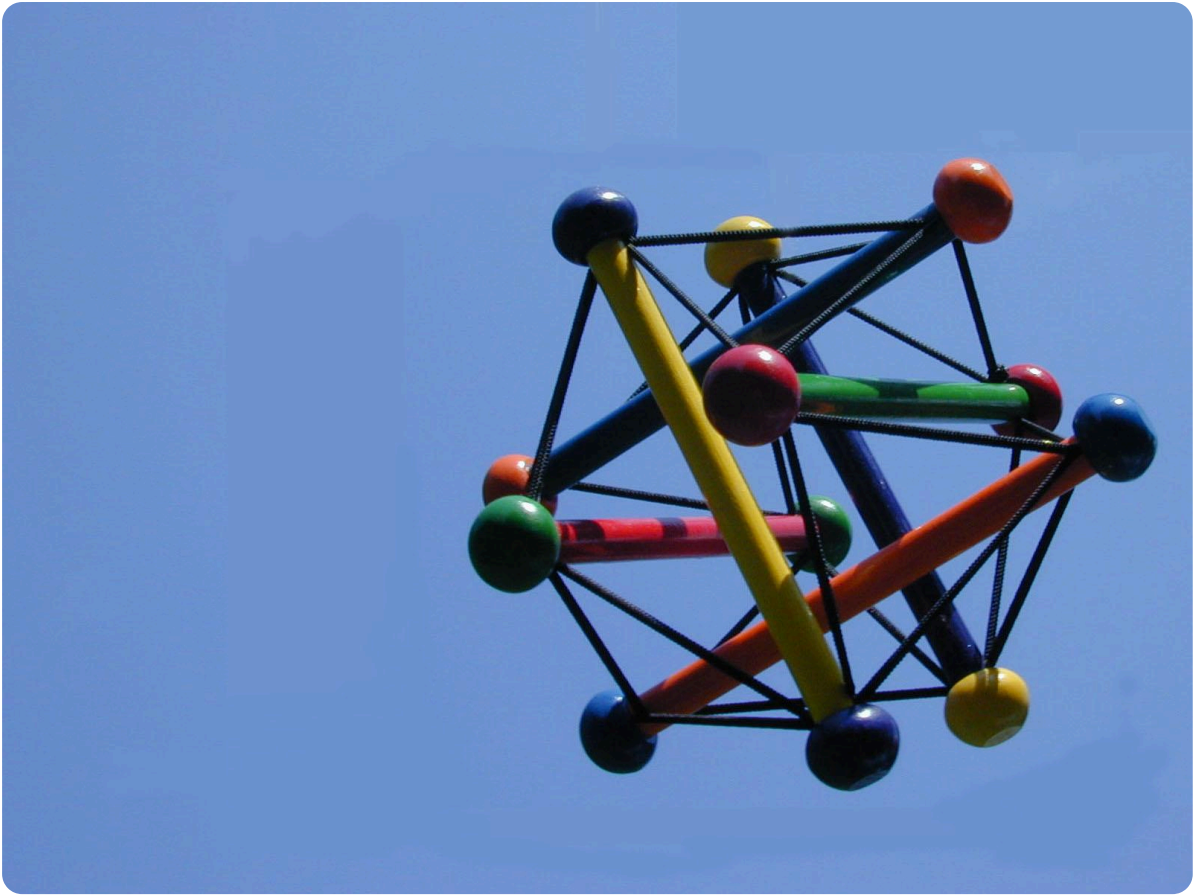


**FIG.** — Protein, Zinc, Sulfate

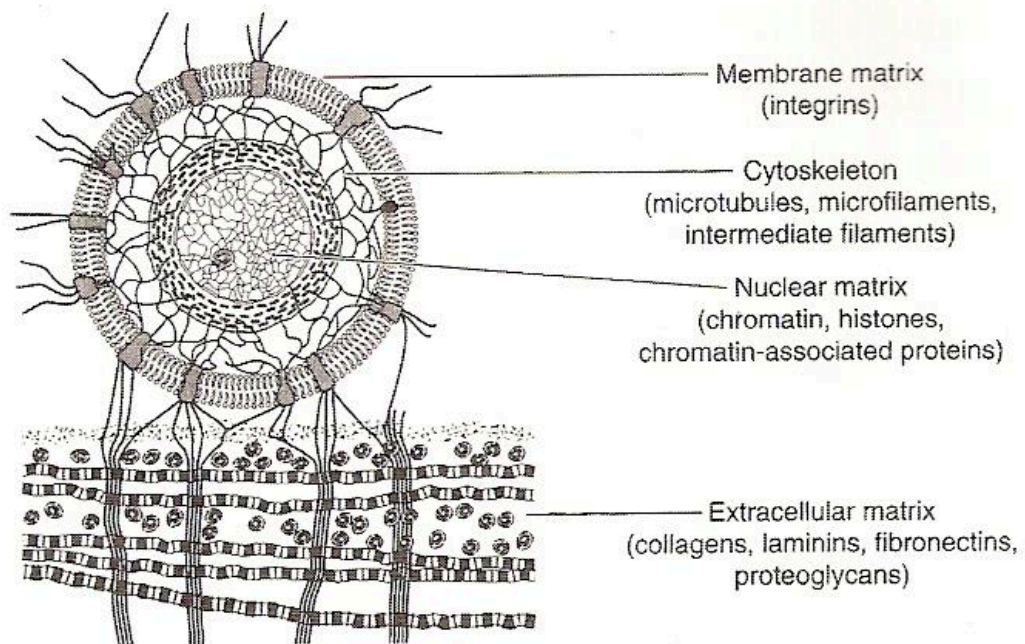


*FIG. — Tensegrity structure*





**FIG.** — *Tensegrity*



4.14 Tissue matrix system as described by Pienta and Coffey, 1991. (Reproduced from Oschman JL (2000). *Energy Medicine*. New York: Churchill Livingstone. Fig. 4.6, p. 66. Reprinted with permission from Elsevier Inc. and Medical Hypotheses.)

**FIG.** — Tissue matrix system